The Overfitting Astronaut (Task-3)

Rules:

My classification rule is: "I will predict YES - Follow Sign if:

(1) The astronaut wears a blue shirt
(2) The astronaut holds a white sign
(3) The sign contains the word "FOLLOW" in black marker.
(4) The sign is not upside down

If all these conditions are met, the prediction will be YES. Otherwise, the prediction will be NO.

Test Rule on New Photos:

Photo A: NO
Reason: The astronaut wears a white EVA suit, and the word "FOLLOW" is written is red marker.

Photo B: NO
Reason: The sign is upside down, so it does not match my rule.

Photo C: NO
Reason: The sign contains the word STOP instead of FOLLOW.

Photo D: NO
Reason: The astronaut wears a white EVA suit and holds an arrow instead of the sign used in my rule.

Compare with the Truth:

Photo	My Prediction	Actual Result
A	NO	YES
B	NO	YES
C	NO	NO
D	NO	YES

Accuracy Calculation:

Correct predictions = 1

Total imeges = 4

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

$$= \frac{1}{4} \times 100$$

$$= 25\%$$

Reflection Questions:

(1) Ans: My rule got 1 out of 4 photos correct. It correctly identified photo C but failed on photos A, B and D.

(2) Ans: My rule focused too much on the blue shirt and specific sign features. This happened because the training data was biased.

(3) Ans: My rule was too specific about the sign's orientation. This is called overfitting because the model memorised unnecessary details instead of learning the general concept.

(4) Ans: I would collect photos with different astronaut suits, sign designs, colours, orientations, and lighting conditions.

(5) Ans: The model fail to recognise astronauts wearing white EVA suits, I would create a diverse training set with different astronaut suits, backgrounds, lighting conditions, and sign designs.

Conclusion:

This task helped me understand supervised learning, overfitting, data bias, and generalization. I learned that a good machine learning model should focus on important features instead of memorising unnecessary details. A diverse training dataset helps a model perform better on new images.